BCA (Semester-I) Examination, 2022

(Session : 2022-25)

COMPUTER APPLICATION

| Paper Code : BCA-101 |

(Mathematical Foundation)

Time : Three Hours [Maximum Marks : 80]

Note: *Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.*

1. Find the Eigen value and Eigen vector:

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

2. If $y^{1/m} + y^{-1/m} = 2x$, prove that

$$(x^2 - 1)y_2 + xy_1 - m^2y = 0$$

3. Evaluate $\int_0^2 \int_0^x (x^2 + y^2)dydx$

4. If $A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$, $\quad B = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$, $\quad C = \begin{bmatrix} x+y & 8 \\ 0 & x-y \end{bmatrix}$ and $A + B = C$, find the value of x and y.

5. Solve: $\left(x^2 - y^2\right)\frac{dy}{dx} = 2xy$

6. Integrate:

$$\int_0^{\frac{\pi}{2}} \sqrt{1 + \sin x}dx$$

7. Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 3 & 0 \\ 0 & 3 & 4 \end{bmatrix}$$

8. Apply Maclaurin's theorem to obtain the expansion of

$$\log(1 + \tan x)$$

9. Solve: $\frac{d^2y}{dx^2} + y = \sin 2x$

10. Solve: $\frac{d^2y}{dx^2} + \frac{3dy}{dx} + 2y = e^{2x}$

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